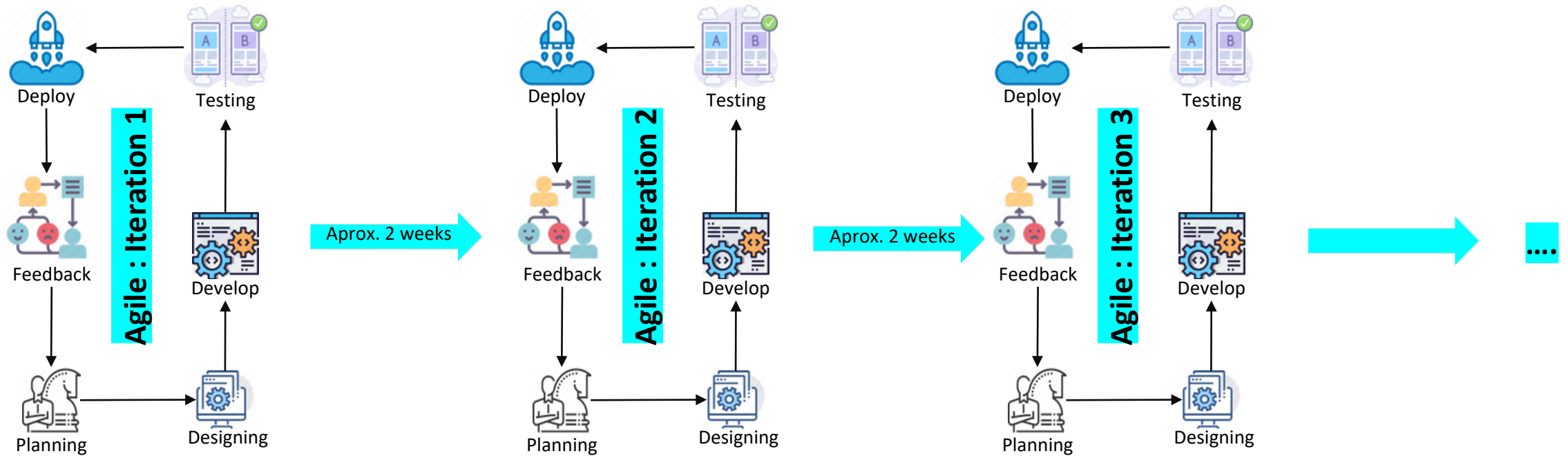


Software Development Life Cycle (SDLC)

Seems this is a perfect model for software development.

then why we need DevOps ?



Software Development Life Cycle (SDLC)

Agile Model:

Disadvantages:



1. Developer computer is used mainly for testing of those features



2. Test and production environment configuration mismatch



3. Usually not tested in production environment

4. Developers team and operations team are in silos

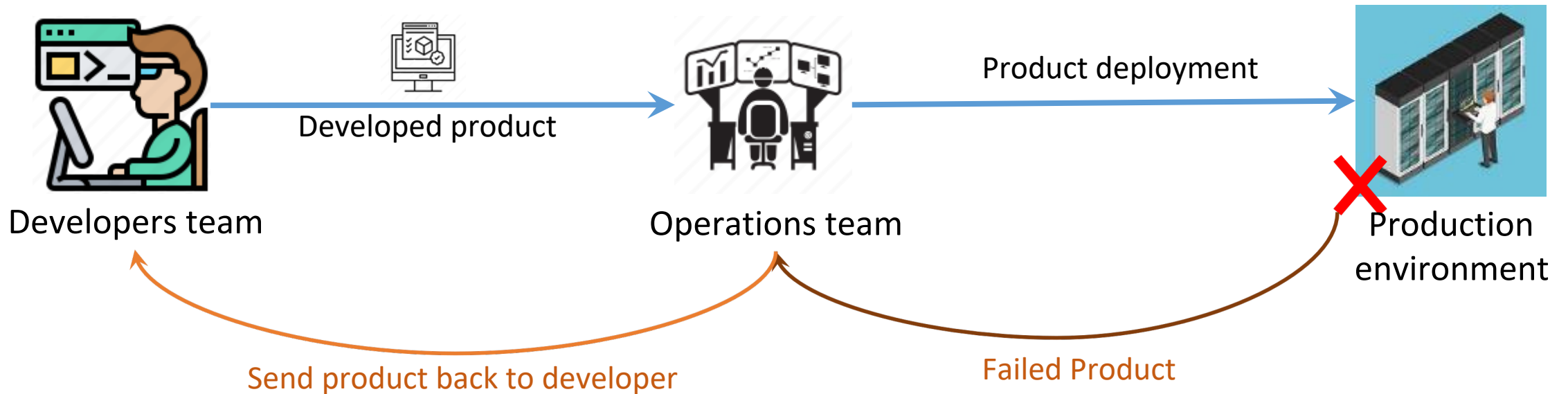


5. Continuous involvement of all the stakeholders.

Software Development Life Cycle (SDLC)

Agile Model:

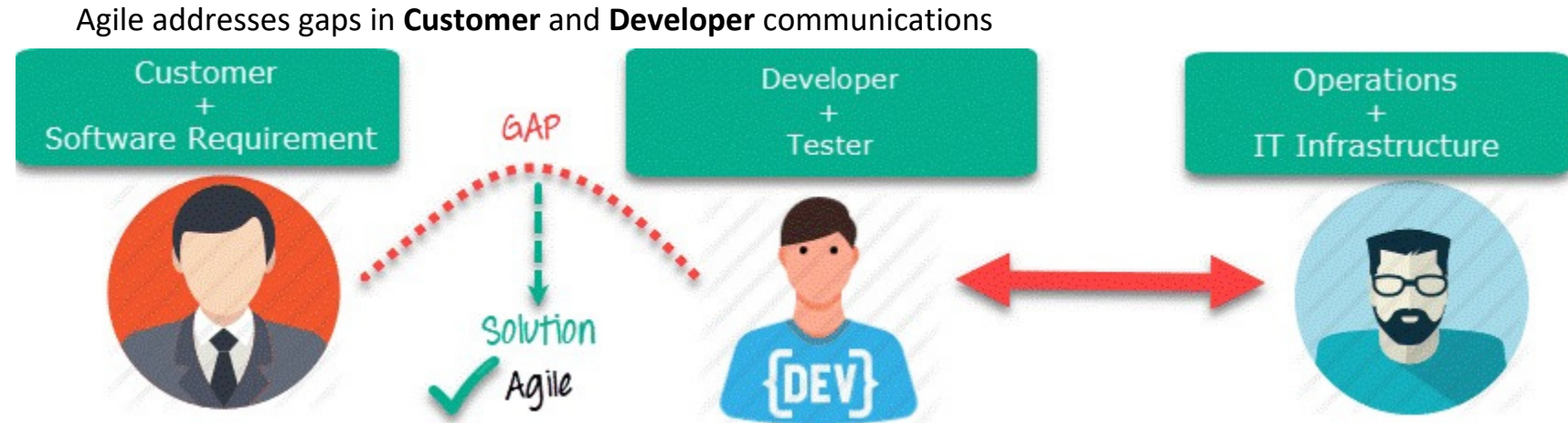
Disadvantages:



Software Development Life Cycle (SDLC)

Agile Model:

Disadvantages:



DevOps addresses gaps in **Developer** and **IT Operations** communications



DevOps

Development + Operations

A culture

DevOps History

- Concept of DevOps emerged out of a discussion between *Andrew Clay* and *Patrick Debois* in 2008. Read more at:
<https://www.appknox.com/blog/history-of-devops>
- Devops days event:
 - a world wide series of technical conferences Topics: software development, IT infrastructure operations, and the intersection between them.
 - Started 2009
 - A series of event in this year (<https://devopsdays.org/events>)

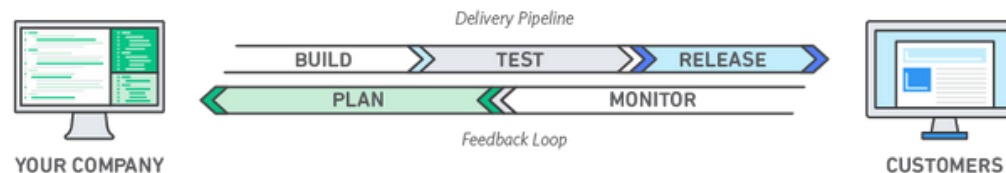
DevOps stats

Netflix:

- 1000s of daily production changes
- 10000s of Virtual instances
- 100,000s of customer interactions/sec
- Billions of time-series metrics
- 100s of operation engineers

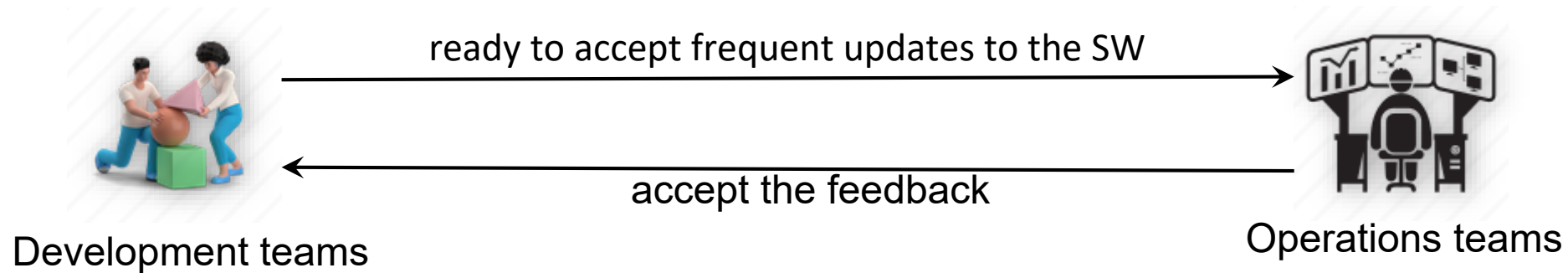
What is DevOps ?

- DevOps =Development+ Operation
- Evolution of Agile software development Model DevOps is not a new technology or a product. DevOps is a ***cultural and professional movement that stresses communication, collaboration and integration between software developers and IT operations professionals.***
- Seeks
 - stability
 - Performance
- Speeds up software deliveries to the business.



How DevOps Works?

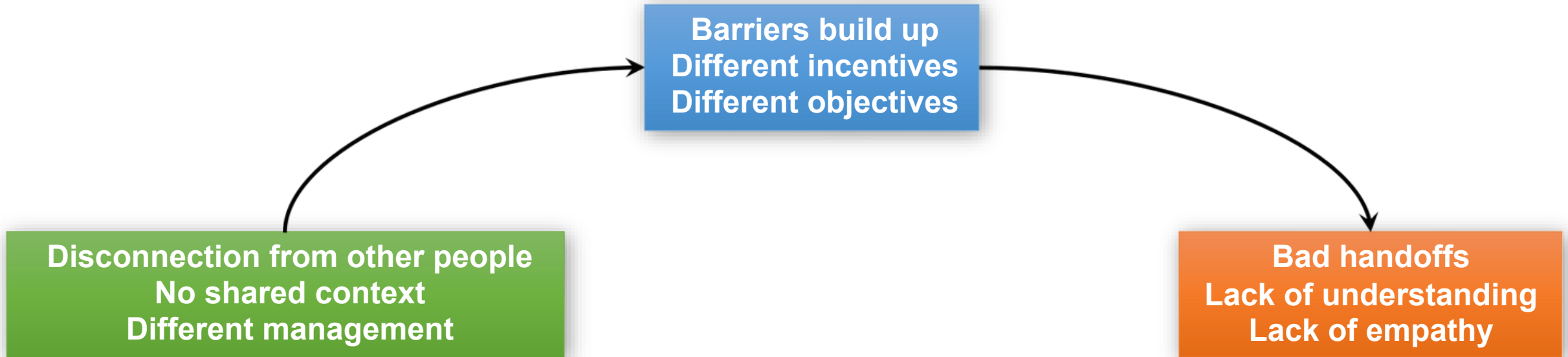
- Development and Operation teams are no longer **siloesd**(isolated).
- Typically the dev-to-ops ratios was between 6:1 and 8:1, now it is around 25:1 [7]



What is a silo?

Development and Operation teams are no longer siloed (isolated).

What is a silo?

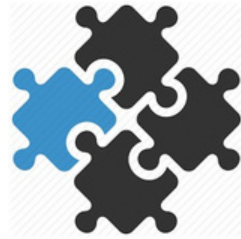


What is a silo?

How to bring Development and Operation teams to one place ?



Communication



Integration



Collaboration

DevOps Goals

- Break down silos
- Create cross-functional teams
- Faster, smaller, more frequent releases
- Improved
 - Time to market
 - Quality of code, products and services
 - Productivity
 - Customer satisfaction
 - Employee satisfaction
- Less waste and fewer defects
- Lower long-term costs

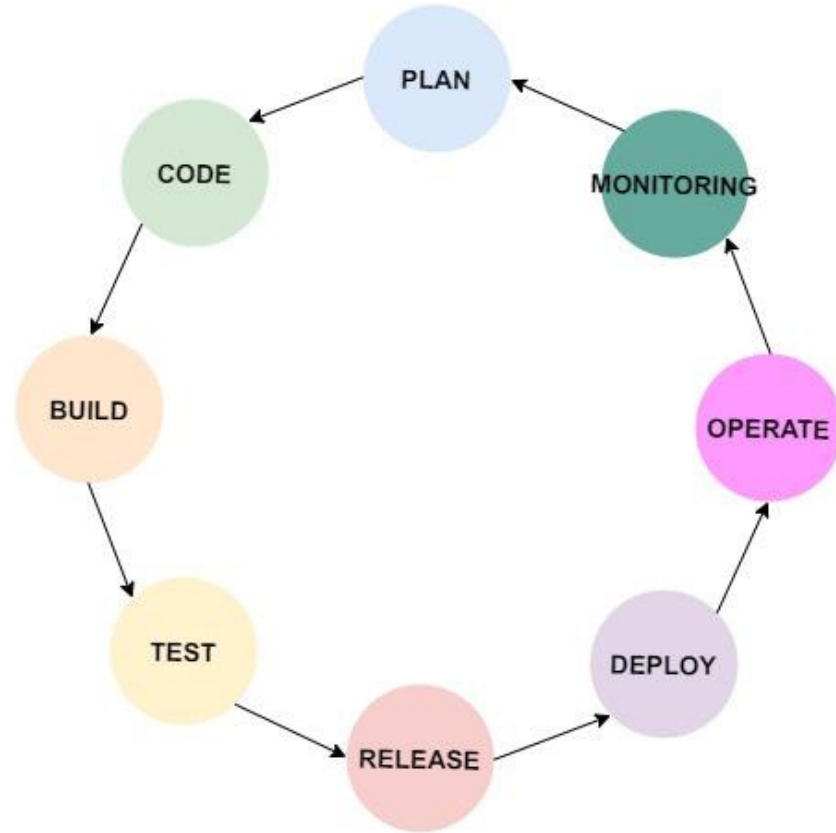
Rule for DevOps Success

- Stay away from “They” culture
- Welcome “We” culture
- Do not play the blame game, take ownership and share resource
 - No punishment
 - Encourage to take risk -> innovation
- Find ways to collaborate – involve others early
- Find ways to automate and make self-service
- Fail fast, continually improve
- Metrics are your friend—always be with measurable outcomes

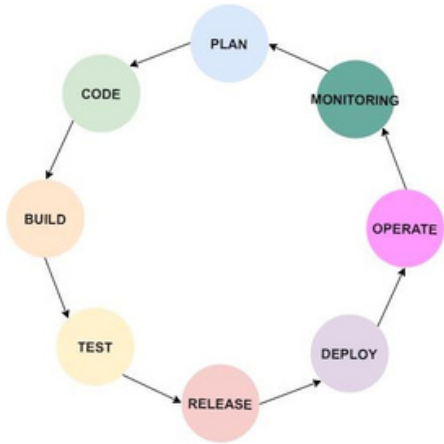
Worth of reading ...

- Leite, Leonardo, et al. "A survey of DevOps concepts and challenges." *ACM Computing Surveys (CSUR)* 52.6 (2019): 1-35.
- Ebert, Christof, et al. "DevOps." *Ieee Software* 33.3 (2016): 94-100
- How to Deploy DevOps in an Organization
- How to Re-design Systems Toward Continuous Delivery
- How to Assess the Quality of DevOps Practices in Organizations
- Many more.....

DevOps Phases



DevOps Phases-Plan



Tools:

- Jira Software
- Confluence
- Slack

Need to have a Vision -> goal for all participants

Idea -> Planning -> Strategy -> Success

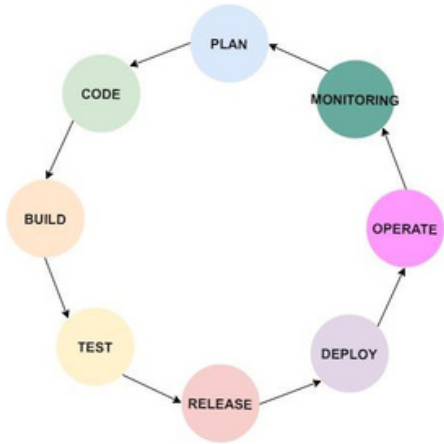
What are the thing that should be in the agenda/mind:

- customer, client,
- software requirement,
- enhancement request, bug fix, line-of-business,
- features, idea

Break work down into smaller, manageable chunks for quicker deployments

Addresses changes and evolutions according to a continuous improvement process and continuous feedback.

DevOps Phases-Code



Here you mainly

- write code

Use of plugins to

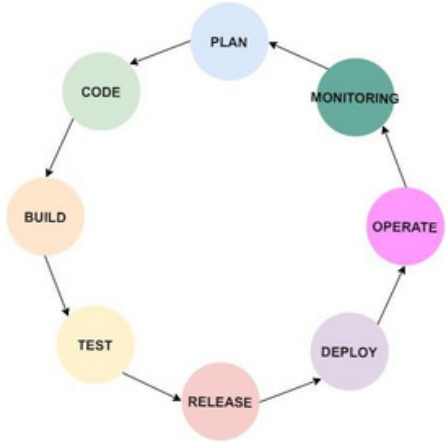
- consistent code-styling
- avoid common security flaws
- Avoid code anti-patterns (e.g. no proper modularize, one solution for all problem, dead code – not sure what it does, unnecessary code/modules/classes, etc).

Several VCS for maintaining the

- codes designing
- infrastructure,
- automating processes,
- defining tests

We will discuss **VCS** in another lecture.

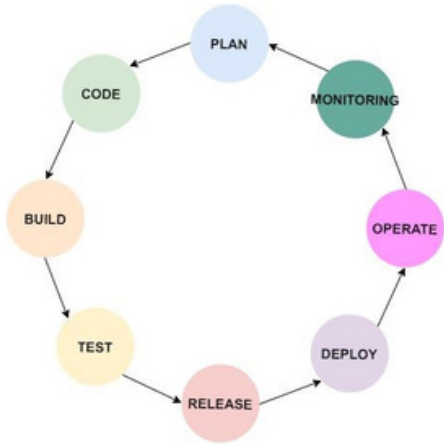
DevOps Phases-Build



Write code ->

1. Commit to the code repo
2. create pull request
3. code review: consistent code-styling , avoid common security flaws , Avoid code anti-patterns.
4. automated tests for possible bugs
5. Merge pull request

DevOps Phases-Test



Tools:

- Selenium
- TestNG
- JUnit

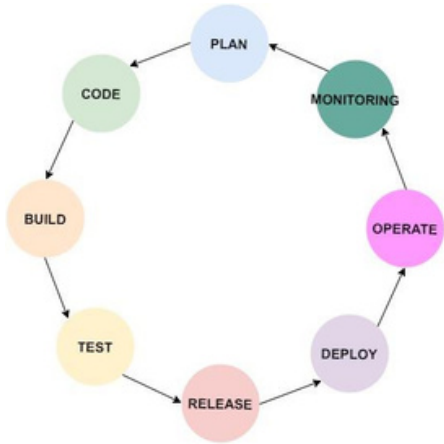
A test can be:

- Scripted (e.g. test case or automated test) or exploratory
- scenario based testing
- Executed manually or through automation
- Linked to/cover one or more requirements
- Like any other Jira issue type, it can be labeled, prioritized, assigned to components, and commented on
- Use of test environment

Types of tests:

- unit tests,
- integration tests,
- functional tests,
- end-to-end tests,
- acceptance tests,
- performance tests.

DevOps Phases-Release



Codesuccessfully passesthrough all test phases (unit/integration/system tests)

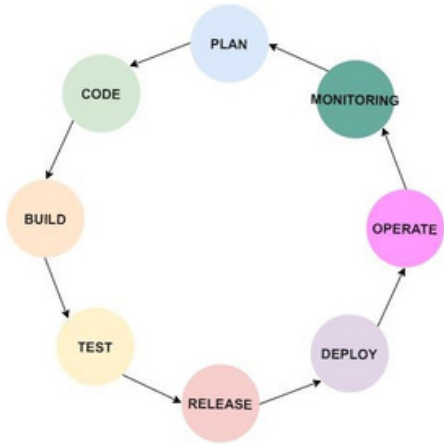
- Build is ready and can be released
- Release can be made automated or manual

COMPANY	DEPLOY FREQUENCY	DEPLOY LEAD TIME	RELIABILITY	CUSTOMER FEEDBACK
AMAZON	23.000/day	minutes	high	high
GOOGLE	5.500/day	minutes	high	high
NETFLIX	500/day	minutes	high	high
FACEBOOK	1/day	hours	high	high
TWITTER	3/week	hours	high	high
TYPICAL ENTERPRISE	once every 9 months	months or more	low/medium	low/medium

From "The Phoenix Project"

Src: <https://imgur.com/a/3uBZKBN>

DevOps Phases-Deploy



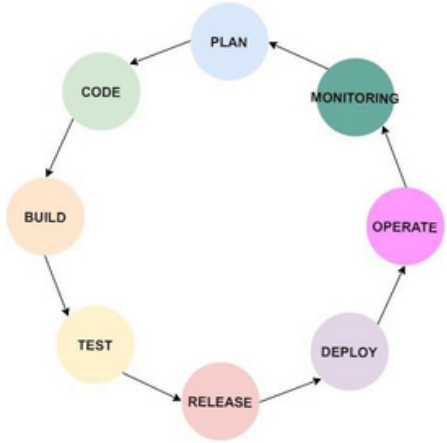
You basically deploy the product on a production environment

Product is ready for

- different deployment contexts
- different deployment environments
- different deployment platforms

If the **release process** for deployment to the production environment is also automated, it is referred to as continuous deployment.

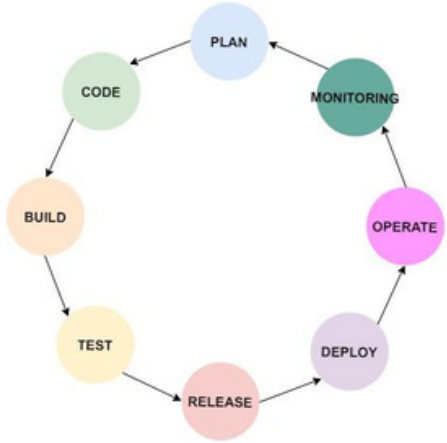
DevOps Phases-Operate



The product live in production environment Operations team relies on automated tools wherever possible for

- configuration management
- scaling
- load balancing

DevOps Phases-Monitor



- Monitor the environments
 - Capturing metrics describing the software's performance.
 - Collects customers' feedback on their experiences
 - analytics on customer behavior performance, errors and
 - more
- Monitor the DevOps pipeline itself

Continuous everything culture in DevOps

